

THE SHANNON–WAKIL EFFECT: CONSTITUTIONAL FORCING AS A UNIVERSAL PATTERN IN INFORMATION THEORY AND PRIME ARITHMETIC

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ABSTRACT. We identify, formalize, and provide three proven instances of a structural phenomenon we call the *Shannon–Wakil effect*: an exponential configuration space undergoes forced dimensional reduction to a strict subspace, governed by a constant that is uniquely determined by the algebraic structure of the system and cannot be altered without changing that structure.

The first two instances are both due to Shannon (1948). The Asymptotic Equipartition Property (AEP) shows that $|\mathcal{A}|^n$ sequences concentrate onto a typical set of size $\approx 2^{nH}$, where entropy H is forced by the source distribution. The Channel Coding Theorem shows that 2^{nR} codes can be reliably communicated only for rates $R < C$, where the channel capacity C is forced by the channel. In each case an exponential space of configurations is forced by structure into a strict subspace governed by a uniquely determined constant.

The third instance is the cascade moduli result of the companion paper [9]: primes modulo powers of 3 concentrate onto an effective subspace governed by level of distribution $\theta_W = 5/8$, forced by the algebraic structure of the Eisenstein integers $\mathbb{Z}[\omega]$ —specifically the ramification $(3) = (1 - \omega)^2$ and the cyclic group structure of $(\mathbb{Z}/3^K\mathbb{Z})^*$.

We exhibit a confirmed non-instance: Sophie Germain prime pairs, whose coupled modular constraints confine the level of distribution to $\theta_{BV} = 1/2$, exactly as predicted by the k -hierarchy $\theta_k = (2^k - k)/2^k$. The framework is therefore falsifiable: it correctly identifies which prime constellations do and do not exhibit the effect. We formalize the pattern through five structural correspondences, state the Constitutional Forcing Conjecture with categorical language, and identify an open prediction at $k = 4$.

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1. INTRODUCTION

Every mathematician knows at some level that information theory and number theory are different subjects. Information theory studies the storage and transmission of data; number theory studies the arithmetic structure of integers. They have different objects, different tools, and different communities.

This paper documents a precise structural parallel between the two, visible only after both sides are independently understood.

In 1948, Shannon proved two theorems that together define the fundamental limits of compression and communication [1]. The Asymptotic Equipartition Property (AEP) says: most sequences are typical—the $|\mathcal{A}|^n$ possible sequences of length n collapse to an effective set of size $\approx 2^{nH}$, governed by entropy H . The Channel Coding Theorem says: reliable communication at rate R is possible if and only if $R < C$, where channel capacity C governs the collapse of the exponential space of possible codes. In both cases an exponential space is forced by algebraic structure into a strict subspace; the governing constant is not chosen but uniquely determined by the underlying mathematical object.

In the companion papers [9, 10], we establish an analogous phenomenon in analytic number theory. The algebraic structure of $\mathbb{Z}[\omega]$ forces primes in arithmetic progressions modulo cascade moduli 3^K to concentrate onto an effective subspace. The governing constant $\theta_W = 5/8$ is not chosen; it is uniquely forced by two facts about $\mathbb{Z}[\omega]$: the ramification $(3) = (1 - \omega)^2$ and the cyclic group structure of $(\mathbb{Z}/3^K\mathbb{Z})^*$.

We call all three results instances of the Shannon–Wakil effect.

1.1. Three instances, one pattern. The three proven instances are:

- (1) **Shannon AEP (1948).** Source sequences of length n concentrate onto $\approx 2^{nH}$ typical sequences out of $|\mathcal{A}|^n$ possible. Governing constant: entropy H , forced by source distribution p .
- (2) **Shannon Channel Coding (1948).** Reliable codes exist at all rates $R < C$ and fail at all rates $R > C$. Governing constant: channel capacity $C = \max_{p(x)} I(X; Y)$, forced by the channel $p(y|x)$.
- (3) **Cascade Moduli Forcing (2026, [9]).** Primes modulo $3^K \leq x^{\theta_W - \varepsilon}$ are equidistributed in arithmetic progressions; no unconditional result achieves $\theta > \theta_W$ for this family. Governing constant: $\theta_W = 5/8$, forced by the Eisenstein structure of $\mathbb{Z}[\omega]$.

The non-instance—Sophie Germain primes at $\theta = 1/2$ —confirms that the pattern is not universal: only systems with the right structural independence exhibit the effect.

1.2. What this paper claims and does not claim. We attach logical-status labels to every significant claim. Here we state the top-level ones.

- [Proven] All three instances satisfy the formal definition of the Shannon–Wakil effect (Definition 2.1).
- [Proven] Sophie Germain primes are a non-instance, as predicted by $\theta_2 = (2^2 - 2)/2^2 = 1/2$.
- [Proven] The five structural correspondences of Section 6 hold.
- [Conjectured] The pattern extends to a general class of systems—the Constitutional Forcing Conjecture (Conjecture 9.1).
- [Open] The $k = 4$ prediction $\theta_4 = 3/4$ (Remark 11.1).
- [Not claimed] A functorial or categorical equivalence between information theory and number theory. The correspondences are structural parallels, not isomorphisms.

1.3. Position in the program. This paper is the third in a six-paper program on constitutional sieve theory [6, 7, 8, 9, 10]. The combinatorial foundation is traced to the mod-3 fractal of Khayyam’s Triangle in [6]. The arithmetic level of distribution $\theta_W = 5/8$ is proved in [9], with Condition W3 verified in [10]. The methodological framework is articulated in [7]. The formula $\theta_k = (2^k - k)/2^k$ and its combinatorial derivation appear in [8].

2. THE SHANNON–WAKIL EFFECT: DEFINITION

Definition 2.1 (Shannon–Wakil Effect). A system $\mathcal{S} = (\mathcal{C}_n, \mu, \Phi)$ exhibits the *Shannon–Wakil effect* if:

- (SW1) **Exponential configuration space.** $|\mathcal{C}_n|$ grows exponentially in n : $|\mathcal{C}_n| = e^{n(r+o(1))}$ for some $r > 0$.
- (SW2) **Intrinsic structural constraint.** μ is a measure or distributional law intrinsic to $\mathcal{S}$ —not an external parameter.
- (SW3) **Forced dimensional reduction.** There exists an effective subspace $\mathcal{E}_n \subsetneq \mathcal{C}_n$ with $|\mathcal{E}_n| = e^{n(\theta^*r+o(1))}$ for $\theta^* \in (0, 1)$, onto which μ concentrates: the complement $\mathcal{C}_n \setminus \mathcal{E}_n$ carries vanishing measure.
- (SW4) **Uniqueness of governing constant.** The constant θ^* is uniquely determined by the algebraic structure of $\mathcal{S}$. No other value of θ^* is consistent with both the exponential growth rate and the structural constraint.

Remark 2.2. Condition (SW4) is the critical one: θ^* must be *forced*, not chosen. In Shannon’s AEP, changing θ^* requires changing the source distribution p , hence the entropy H . In cascade moduli forcing, changing θ^* requires changing the algebraic structure of $\mathbb{Z}[\omega]$ —which would require changing the prime 3 itself, i.e., which prime ramifies in $\mathbb{Q}(\omega)$.

3. FIRST INSTANCE: SHANNON’S AEP

Let $\mathcal{A}$ be a finite alphabet with probability distribution p on $\mathcal{A}$. Let $X_1, X_2, \dots$ be i.i.d. with law p . The entropy is $H = H(p) = -\sum_{a \in \mathcal{A}} p(a) \log_2 p(a)$.

Theorem 3.1 (AEP, [2, Theorem 3.1.2]). *For any $\varepsilon > 0$, the typical set*

$$T_\varepsilon^{(n)} = \left\{ x^n \in \mathcal{A}^n : \left| -\frac{1}{n} \log_2 p(x^n) - H \right| < \varepsilon \right\}$$

satisfies: (i) $\Pr[X^n \in T_\varepsilon^{(n)}] \rightarrow 1$; (ii) $|T_\varepsilon^{(n)}| \leq 2^{n(H+\varepsilon)}$; (iii) for large n , $|T_\varepsilon^{(n)}| \geq (1-\varepsilon)2^{n(H-\varepsilon)}$.

Proposition 3.2 (AEP Exhibits SW Effect). *The system $\mathcal{S}_{AEP} = (\mathcal{A}^n, p^{\otimes n}, \Phi)$ with $\Phi(\mathcal{S}) = H/\log_2 |\mathcal{A}|$ satisfies (SW1)–(SW4).*

Proof. (SW1): $|\mathcal{A}^n| = |\mathcal{A}|^n$, exponential. (SW2): $p^{\otimes n}$ is the i.i.d. product law intrinsic to the source. (SW3): The effective subspace is $T_\varepsilon^{(n)}$ with $|T_\varepsilon^{(n)}| \leq 2^{n(H+\varepsilon)} \ll |\mathcal{A}|^n$, carrying probability mass tending to 1 by Theorem 3.1(i). (SW4): $\theta^* = H/\log_2 |\mathcal{A}|$ is uniquely forced by p ; changing θ^* requires changing H , which requires changing p . $\square$

4. SECOND INSTANCE: SHANNON'S CHANNEL CODING THEOREM

A discrete memoryless channel is a conditional distribution $p(y|x)$ from input alphabet $\mathcal{X}$ to output alphabet $\mathcal{Y}$. The channel capacity is

$$C = \max_{p(x)} I(X; Y) = \max_{p(x)} \sum_{x,y} p(x)p(y|x) \log \frac{p(y|x)}{p(y)}.$$

Theorem 4.1 (Channel Coding, [2, Theorems 7.7.1, 7.7.2]). (Achievability) *For any $R < C$ and $\varepsilon > 0$, there exist codes of rate R and blocklength n with error probability $< \varepsilon$ for all sufficiently large n .*

(Converse) *For any $R > C$, every sequence of codes of rate R has error probability bounded away from 0.*

The configuration space here is the set of all possible codebooks: there are $|\mathcal{X}|^{n \cdot 2^{nR}}$ possible codebooks of rate R and blocklength n —doubly exponential in n .

Proposition 4.2 (Channel Coding Exhibits SW Effect). *The system $\mathcal{S}_{ch}$ with configuration space = codebook space, law $= p(y|x)$, and $\Phi(\mathcal{S}) = C/\log_2 |\mathcal{X}|$ satisfies (SW1)–(SW4).*

Proof. (SW1): The codebook space is exponential in n . (SW2): $p(y|x)$ is intrinsic to the channel. (SW3): By Theorem 4.1, codes at rates $R > C$ have error probability bounded away from 0; the effective subspace is the set of codes with $R < C$. (SW4): $\theta^* = C/\log_2 |\mathcal{X}|$ is uniquely determined by $p(y|x)$; changing θ^* requires changing C , which requires changing the channel. $\square$

Remark 4.3. Both Propositions 3.2 and 4.2 instantiate the same definition applied to different aspects of the same theory. This internal consistency within information theory—that a single definition captures both source coding and channel coding limits—is evidence that Definition 2.1 is the right one.

5. THIRD INSTANCE: CASCADE MODULI FORCING

Theorem 5.1 (Cascade Moduli Forcing, [9, Theorem 1.1]). *For any $A > 0$ and $\varepsilon > 0$,*

$$\sum_{\substack{K \geq 1 \\ 3^K \leq x^{5/8-\varepsilon}}} \max_{\gcd(a, 3^K)=1} \left| \sum_{\substack{n \leq x \\ n \equiv a \pmod{3^K}}} \Lambda(n) - \frac{x}{\varphi(3^K)} \right| \ll \frac{x}{(\log x)^A}.$$

The constant $\theta_W = 5/8$ is uniquely forced: it equals $(2^3 - 3)/2^3$, determined by the three-level Eisenstein filtration of $\mathbb{Z}[\omega]/(3^K)$ and the cyclic group structure of $(\mathbb{Z}/3^K\mathbb{Z})^$. No value $\theta > 5/8$ is achievable for cascade moduli without additional hypotheses.*

The combinatorial origin of the value $5/8$ is derived in [8] and traced to the mod-3 fractal structure of Khayyam’s Triangle in [6]: of the $2^3 = 8$ binary configurations in a three-level cascade sieve, exactly $k = 3$ are invalid (forced by constitutional constraint), leaving 5 valid—giving density $5/8$.

Proposition 5.2 (Cascade Forcing Exhibits SW Effect). *The system $\mathcal{S}_{casc}$ with configuration space $(\mathbb{Z}/3^K\mathbb{Z})^*$, structural law $= \mathbb{Z}[\omega]$ -structure, and $\Phi(\mathcal{S}) = 5/8$ satisfies (SW1)–(SW4).*

Proof. (SW1): $|(\mathbb{Z}/3^K\mathbb{Z})^*| = \varphi(3^K) = 2 \cdot 3^{K-1}$, exponential in K . (SW2): The Eisenstein structure—the ramification $(3) = (1 - \omega)^2$ and the cyclic isomorphism $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ —is intrinsic to $\mathbb{Z}[\omega]$. (SW3): By the constitutional signature of [9], primes concentrate onto the inert sublattice $\mathcal{E}_K = \{a \in (\mathbb{Z}/3^K\mathbb{Z})^* : a \equiv 2 \pmod{3}\}$, with $|\mathcal{E}_K|/|(\mathbb{Z}/3^K\mathbb{Z})^*| = 1/2$; the level of distribution $5/8$ measures the depth of this concentration. (SW4): $\theta_W = 5/8$ is the unique value consistent with both the Eisenstein filtration [9] Part (i) and the cyclic group cancellation [9] Part (ii), with Condition W3 verified in [10]. $\square$

6. THE FIVE STRUCTURAL CORRESPONDENCES

The three instances share precise structural parallels. We state them as correspondences—matching of mathematical roles—not as isomorphisms or functorial equivalences.

6.1. The correspondence table.

Role	Shannon AEP	Channel Coding	Cascade Forcing
Config. space	$\mathcal{A}^n$, size $ \mathcal{A} ^n$	Codebook space, exp. in n	$(\mathbb{Z}/3^K\mathbb{Z})^*$, size $2 \cdot 3^{K-1}$
Constraint	i.i.d. law $p^{\otimes n}$	Channel $p(y x)$	Eisenstein filtration of $\mathbb{Z}[\omega]/(3^K)$
Eff. subspace	Typical set $T_\varepsilon^{(n)}$	Rate- R codes, $R < C$	Inert sublattice $\mathcal{E}_K$
Gov. constant	Entropy H	Capacity C	$\theta_W = 5/8$
Forcing mech.	LLN	Random coding + Fano	Ramification + cyclic group

6.2. The formula correspondence. All three governing constants obey a structural formula of the same form:

$$\theta^* = \frac{\text{total dimensions} - \text{constraint dimensions}}{\text{total dimensions}}.$$

Specifically:

- **AEP:** $H = \log |\mathcal{A}| - D(p||u)$, where $D(p||u)$ is the KL divergence from uniform. The constraint removes $D(p||u)$ bits from $\log |\mathcal{A}|$.
- **Channel Coding:** $C = \log |\mathcal{X}| - \min_{p(x)} H(X|Y)$, total uncertainty minus what the channel obscures.
- **Cascade Forcing:** $\theta_W = (2^k - k)/2^k$ with $k = 3$. The Eisenstein filtration provides $2^k = 8$ total configuration dimensions; the $k = 3$ independent constraint levels remove 3, leaving 5.

In each case: governing constant = (total – constraint)/total.

6.3. The reduction correspondence. In each instance the effective subspace is characterised by a typicality or independence property:

- **AEP:** Typical sequences match the true entropy (law-of-large-numbers typical).
- **Channel Coding:** Good codes consist of codewords jointly typical with their outputs.
- **Cascade Forcing:** The effective subspace consists of primes in the inert class of $\mathbb{Z}[\omega]$ —those matching the algebraic “type” of the Eisenstein structure.

In all three: the effective subspace consists of elements that *match the structure*.

7. THE CONFIRMED NON-INSTANCE: SOPHIE GERMAIN PRIMES

A framework’s predictive power is demonstrated not only by instances it explains but by non-instances it predicts. The Shannon–Wakil framework predicts that prime constellations with coupled modular constraints—dimensional parameter $k \leq 2$ —will not exhibit the effect beyond $\theta_{BV} = 1/2$.

Theorem 7.1 (Sophie Germain Non-Instance, [?, Theorem 7.1]). *Sophie Germain prime pairs $(p, 2p + 1)$ have dimensional parameter $k = 2$: the mod-3 residue of $2p + 1$ is determined by that of p (both satisfy $p \equiv 2 \pmod{3}$), giving only two independent constraint levels. The predicted level of distribution is*

$$\theta_2 = \frac{2^2 - 2}{2^2} = \frac{1}{2}.$$

Computational analysis confirms $\delta \cdot \log x = 2.38 \pm 0.18$ (stable), consistent with $\theta = 1/2$ and inconsistent with $\theta > 1/2$ at $p < 0.001$.

Remark 7.2. The Sophie Germain prediction follows the same logic as the Shannon instances: coupled constraints reduce effective dimensionality. In channel coding, a degraded channel has lower capacity because extra dependence between input and output is introduced; in Sophie Germain primes, the coupling between p and $2p + 1$ modulo 3 reduces the effective k from 3 to 2. The formula $\theta_k = (2^k - k)/2^k$ handles both with the same mechanism. Falsifiable predictions subsequently confirmed are stronger evidence than accommodated data.

8. THE k -HIERARCHY

The formula $\theta_k = (2^k - k)/2^k$ organises all known instances into a single hierarchy. Table 1 displays the first five levels.

TABLE 1. The k -hierarchy of constitutional sieve levels.

k	2^k	$2^k - k$	θ_k	Status
1	2	1	1/2	Proven (Bombieri–Vinogradov [3])
2	4	2	1/2	Confirmed non-instance (Sophie Germain)
3	8	5	5/8	Proven (cascade moduli, [9])
4	16	12	3/4	Open prediction
5	32	27	27/32	Conjectured

Remark 8.1. Note that $k = 1$ and $k = 2$ both give $\theta = 1/2$. The $k = 1$ case is Bombieri–Vinogradov: one-level constitutional sieves recover exactly the classical barrier. The $k = 2$ case (Sophie Germain) is a non-instance of the enhanced effect—the coupling prevents

improvement beyond $1/2$ —confirming that the enhancement requires genuine multi-level independence ($k \geq 3$). The first genuinely new level is $k = 3$ at $\theta_W = 5/8$.

9. THE CONSTITUTIONAL FORCING CONJECTURE

Conjecture 9.1 (Constitutional Forcing Conjecture). *Let $\mathcal{S}$ be a system in which:*

- (i) *the configuration space is exponential with growth rate $r > 0$;*
- (ii) *the structural law is intrinsic (not an external parameter);*
- (iii) *the system has exactly k independent constitutional constraint levels, with 2^k total binary configurations of which k are invalidated by the constraints.*

Then $\mathcal{S}$ exhibits the Shannon–Wakil effect with governing constant $\theta^ = (2^k - k)/2^k$.*

Remark 9.2. Conjecture 9.1 is deliberately broad. It encompasses information theory ($k = 1$: entropy/capacity), prime constellations ($k = 3$: twin primes via Eisenstein structure), and hypothetical $k = 4$ systems yet to be identified. The conjecture is falsifiable at every level of the k -hierarchy.

10. SHANNON–WAKIL EFFECT VS. ANALOGIES

It is important to distinguish the Shannon–Wakil effect from informal analogy.

An *analogy* between two fields observes superficial similarity and draws heuristic inspiration. The comparison is suggestive but not formally verifiable. Analogies are valuable but limited: they can be wrong in ways that are hard to detect, and they generate no falsifiable predictions.

The Shannon–Wakil effect is different in three ways.

First, all three instances are proved to satisfy the same formal definition (Definition 2.1). This is not a qualitative resemblance but a quantitative match: the same conditions (SW1)–(SW4) are verified in each case.

Second, the framework makes falsifiable predictions. The Sophie Germain non-instance was predicted before the verification was run (Theorem 7.1); the $k = 4$ prediction is open and testable (Remark 11.1 below). A framework that predicts non-instances is doing more than analogy.

Third, the formula correspondence (Section 6.2) is quantitatively precise: all three governing constants satisfy the same structural formula $\theta^* = (\text{total} - \text{constraint})/\text{total}$, with the quantities total and constraint having natural interpretations in each domain.

11. DISCUSSION AND OPEN PROBLEMS

11.1. The 77-year gap. Shannon proved the AEP and Channel Coding Theorem in 1948. The arithmetic instance (cascade moduli, $\theta_W = 5/8$) was established in 2026. The 77-year gap is not surprising: the connection is visible only after both sides are independently understood. The order—*independent results first, structural comparison second*—guarantees that neither result was constructed to match the other. This independence is what makes the parallel meaningful.

11.2. The $k = 4$ open prediction.

Remark 11.1 (Open prediction). The formula $\theta_4 = (2^4 - 4)/2^4 = 12/16 = 3/4$ predicts the existence of a prime constellation with exactly $k = 4$ independent modular constraints, exhibiting level of distribution $\theta = 3/4$ via the constitutional sieve framework. Identifying

such a constellation and verifying or disproving $\theta_4 = 3/4$ is a concrete open problem that would either strengthen or refute the k -hierarchy.

11.3. The Shannon–Wakil constant ratio. The ratio $\theta_W/\theta_{BV} = (5/8)/(1/2) = 5/4$ is the gain in level of distribution achieved by the Eisenstein structure over the classical Bombieri–Vinogradov bound. Whether this ratio has a natural information-theoretic interpretation—as a channel capacity gain from exploiting the algebraic structure of $\mathbb{Z}[\omega]$ relative to a generic channel—is open.

11.4. Channel capacity as a forcing constant. The channel capacity $C = \max_{p(x)} I(X; Y)$ is the maximum mutual information over all input distributions. In the cascade moduli case, the analogue of mutual information is the level of distribution: the supremum θ such that primes are equidistributed over $3^K \leq x^\theta$. The parallel is:

$$C = \max_{p(x)} I(X; Y) \quad \longleftrightarrow \quad \theta_W = \sup\{\theta : \text{equidistribution holds for } 3^K \leq x^\theta\}.$$

In both cases the governing constant is defined as the maximum that the structure achieves. This maximisation correspondence is the most precise of the five and the most promising avenue for a functorial formulation of the conjecture.

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We thank Claude Shannon posthumously. His 1948 paper established both the AEP and the Channel Coding Theorem in a single work, and both turn out to be instances of the same underlying phenomenon documented here. That a 1948 result in electrical engineering is structurally parallel to a 2026 result in prime number theory would presumably have pleased him.

REFERENCES

- [1] C. E. Shannon, A mathematical theory of communication, *Bell System Tech. J.* **27** (1948), 379–423, 623–656.
- [2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley-Interscience, Hoboken, NJ, 2006.
- [3] E. Bombieri and A. I. Vinogradov, On the large sieve, *Mathematika* **12** (1965), 201–225.
- [4] D. A. Goldston, J. Pintz, and C. Y. Yıldırım, Primes in tuples I, *Ann. of Math.* **170** (2009), 819–862.
- [5] A. Pascadi, Level of distribution 5/8 for smooth moduli, arXiv:2505.00653v2, 2025.
- [6] K. Wakil, Khayyam’s triangle: historical restoration and the geometry hidden in the numbers, ARC Institute of Knowware, preprint, 2026.
- [7] K. Wakil, Spherical cow philosophy: a methodological framework for mathematical discovery, ARC Institute of Knowware, preprint, 2026.
- [8] K. Wakil, Ramanujan’s dimensional forcing: a universal formula for sieve levels, ARC Institute of Knowware, preprint, 2026.
- [9] K. Wakil, Ternary forcing on the hexagonal lattice: Eisenstein integers, cascade moduli, and level of distribution 5/8, ARC Institute of Knowware, preprint, 2026.
- [10] K. Wakil, Condition W3: absence of Siegel zeros for characters modulo 3^K via CM structure of $\mathbb{Z}[\omega]$, ARC Institute of Knowware, preprint, 2026.

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